

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓	✓		✓
CO 4		✓		✓
CO 5		✓		✓
CO 6			✓	✓

Programme	BSc Statistics					
Course Code	ACT1MN102 (P)					
Course Title	Financial Mathematics					
Type of Course	Minor					
Semester	I					
Academic Level	100 - 199					
Course Details	Credit	Lecture	Tutorial	Practical	Total	

		per week	per week	per week	Hours
	4	3	-	2	75
Pre-requisites	Basic knowledge of rates of interest, arithmetic skills, Basic computer skills				
Course Summary	The aim of this course is to provide a grounding in the principles of modelling as applied to actuarial work				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Describe the possible aims of data analysis	U	F	Instructor-created exams / Quiz
CO2	Explain the meaning and value of reproducible research and describe the elements required to ensure a data analysis is reproducible.	U	P	Instructor-created exams / Home Assignments
CO3	Describe why and how models are used including, in general terms, the use of models for pricing, reserving and capital modelling.	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	Apply force of interest in various financial transaction.	An	C	Instructor-created exams / Home Assignments
CO5	Describe how a loan may be repaid by regular instalments of interest and capital.	Ap	P	One Minute Reflection Writing assignments
CO6	Solving cash flow models with sample data with spread sheet	Ap	P	Viva-Voce/Practical Assignment
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hours (45 +30)	Marks (70)
I	Data Analysis		8	10
	1	Aims of data analysis- Descriptive, inferential and predictive	2	
	2	The data analysis process	2	

	3	Data Sources, Big data	2	
	4	Data security privacy and regulation , Reproducible research	2	
	Sections from References: Unit 1: 1.1&1.1.1 to 1.1.3 [Ref 1] Unit 2: 1.2 [Ref 1] Unit 3: 1.3&1.3.1[Ref 1] Unit 4: 1.3.2&1.4 [Ref 1]			
II	Principles of actuarial modelling		11	15
	5	Model, How models are used, Modelling – the benefits and limitations	2	
	6	Stochastic and deterministic models, Discrete and continuous state spaces and time sets	2	
	7	Scenario-based and proxy models, Suitability of a model, Short-term and long-term properties of a model	3	
	8	Analysing the output of a model, Sensitivity testing, Communication of the results	2	
	9	Insurance contracts- Pure endowment, Endowment, term, contingent annuity, a car insurance policy, a health cash plan	2	
	Sections from References: Unit 5: 2.1,2.1.2 &2.2 [Ref 1] Unit 6: 2.3,2.4 [Ref 1] Unit 7: 2.5,2.6&2.7 [Ref 1] Unit 8: 2.8,2.9 & 2.10 [Ref 1] Unit 9: 3.1 (3.1.1 to 3.1.6) [Ref 1]			
III	Real and Money interest rates		12	25
	10	Real and Money interest rates	2	
	11	Deflationary conditions, Usefulness of real and money interest rates	2	
	12	Force of interest as a function of time	3	
	13	Relationship to constant force of interest, Applications of force of interest.	2	
	14	Present values of cash-flows	1	
	15	Payment streams, Sudden changes in interest rates	2	
	Sections from References: Unit 10: 6.1 ([Ref 1] Unit 11: 6.2 & 6.3[Ref 1] Unit 12: 5.4 [Ref 1] Unit 13: 5.4.2 [Ref 1] Unit 14: 7.1 [Ref 1] Unit 15: 7.2.2 & 7.2.3 [Ref 1]			

IV	Loan Schedule & Varying annuities		14	20
	16	Introduction, Calculating the capital outstanding	2	
	17	calculating the interest and capital elements	3	
	18	The loan schedule	1	
	19	Instalments payable more frequently than annually.	2	
	20	The consumer credit-APR	2	
	21	Varying annuities- Annual payments	3	
	22	Decreasing payments	1	
	Sections from References: Unit 16: 11.1&11.2 [Ref 1] Unit 17: 11.3 [Ref 1] Unit 18: 11.4[Ref 1] Unit 19: 11.5 [Ref 1] Unit 20: 11.6[Ref 1] Unit 21: 9.1.1[Ref 1] Unit 22: 9.1.3 [Ref 1]			
V	PRACTICUM		30	
	Do practice problems in spreadsheet from any 5 units of the given list and one additional problem decided by the teacher-in-charge, related to the content of the course. Other units listed here may be used as demonstrations of the concepts taught in the course.			
	1	Real and money interest rates		
	2	Present value of cash-flows		
	3	Payment streams		
	4	Relationship to constant force of interest		
	5	Loan schedule –preparation of loan amortization table		
	6	Loan schedule-Capital and interest calculations		
	7	APR		
	8	Flat rate of interest		
Books and References: 12. Institute of Actuaries Act Ed. <i>Study materials</i> CM1 13. McCutcheon, J.J., Scott William (1986): An introduction to Mathematics of Finance 14. Butcher,M.V., Nesbit, Cecil. (1971)Mathematics of compound interest, Ulrich’s Books 15. David Promislow, S. (2014), Fundamentals of Actuarial Mathematics, John wiley& sons. 16. Newton LBowers, et al (1997):Actuarial Mathematics, The Societies of Actuaries, 2nd Ed 17. Shailaja R. Deshmukh- Actuarial Statistics-an introduction using R, 3 rd Ed.				

Mapping of COs with PSOs and POs:

	PSO 1	PSO 2	PSO 3	PSO4	PSO 5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	2	3	1	3	-	-	1	-	2	2	-	-
CO 2	3	2	-	2	1	-	3	1	2	3	-	-
CO 3	1	2	-	2	-	-	2	-	2	-	-	-
CO 4	3	2	-	1	-	-	3	-	2	2	-	-
CO 5	3	2	-	-	-	-	3	-	2	-	-	-
CO 6	1	1	2	-	3	3	2	2	1	-	2	1

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

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